

MAHAN BALOOEI

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RESEARCH INTERESTS

- Protein language models & foundation models
- Machine learning for biological data
- Variant-effect & pathogenicity prediction
- Computational genomics & multi-omics

EDUCATION

University of Bologna — MSc in Bioinformatics

Oct 2024 – Present

MSc Thesis: *Protein Language Models for Human Protein Variant Pathogenicity Prediction*

Supervisors: Castrense Savojardo, Pier Luigi Martelli, Matteo Manfredi

Ongoing thesis on modernizing the E-SNPs&GO framework for missense variant pathogenicity prediction through updated human variant datasets and contemporary protein language models.

Iran University of Medical Sciences — BSc in Medical Laboratory Sciences

Sept 2019 – Jul 2023

Relevant Coursework: Molecular Diagnostics, Immunoassays, Clinical Chemistry, Haematology, Microbiology, Virology.

PUBLICATIONS

Farmani, S.^{*}, Dehghanbanadaki, N.^{*}, **Balooei, M.^{*}**, Mehralitabar, H., Sabaghian, H., & Naderi-Manesh, H. **Mechanism-based targeting of lactate dehydrogenase tetramerization by naturally derived compounds: A survey of molecular docking, molecular dynamics, and QSAR analysis.** *Computers in Biology and Medicine*, 197(Pt B), 111000, 2025.

^{*}These authors contributed equally.

- [Article link](#)

RESEARCH EXPERIENCE

Protein Language Model-Based Signal Peptide Prediction

Apr 2026

MSc Bioinformatics Project – Independently Implemented

- Built an end-to-end pipeline for eukaryotic signal-peptide prediction using high-confidence UniProtKB/Swiss-Prot annotations.
- Applied MMseqs2 clustering to reduce sequence redundancy and construct biologically meaningful train/benchmark splits.
- Compared von Heijne PSWM, biochemical-feature SVM, CNN-BiLSTM, and ESM-2 embedding-based models.
- Achieved $F1 = 0.962$ and $MCC = 0.957$ on a held-out benchmark using the ESM-2-based model.
- Analyzed amino-acid composition, signal-peptide lengths, kingdom-specific bias, and cleavage-site motifs for biological interpretation.
- [GitHub Repository](#)

Protein Domain and Functional Annotation Using HMMER and MMseqs2

Feb – Mar 2026

- Built an automated protein-domain annotation workflow using profile HMMs, HMMER, and MMseqs2 for large-scale sequence classification.
- Processed structure-derived protein alignments to support domain-level annotation and sequence-based functional inference.
- Compared profile-based and sequence-similarity-based classification workflows, achieving MCC = 0.997 and MCC = 0.991, respectively.
- Connected classical homology and profile-HMM methods to modern protein representation-learning problems such as remote homology detection and protein-family classification.
- [GitHub Repository](#)

Computational Epigenomics Analysis of DNA Methylation (Illumina 450K)

Jun 2025 – Jul 2025

- Conducted an R/Bioconductor workflow for Illumina 450K DNA methylation-array analysis.
- Performed quality control, normalization, PCA-based batch-effect assessment, and differential methylation analysis.
- Identified differentially methylated regions (DMRs) and visualized methylation patterns using heatmaps, volcano plots, and exploratory statistical summaries.
- [GitHub Repository](#)

Chest X-Ray Pneumonia Detection via Transfer Learning

Apr – May 2026

- Compared ResNet50, EfficientNetBo, and DenseNet-121 XRV for pneumonia classification under class imbalance.
- Evaluated threshold selection, ensembling, test-time augmentation, and Grad-CAM interpretability; DenseNet-121 XRV achieved the strongest performance under the maximum-MCC strategy.
- [GitHub Repository](#)

SKILLS

Protein Bioinformatics: ESM-2, protein representation learning, HMMER, MMseqs2, BLAST, UniProt, PDB, protein functional/domain annotation, sequence-function prediction.

Machine Learning & Deep Learning: PyTorch, scikit-learn, TensorFlow, SVMs, CNN-BiLSTM, transfer learning, representation learning, model evaluation, class-imbalance handling.

Functional Genomics & Epigenomics: R/Bioconductor, minfi, Illumina 450K methylation analysis, differential methylation, PCA-based batch-effect assessment.

Research Computing: Python, R, Bash/Linux, SQL, Git/GitHub, Docker, HPC/Slurm, conda, Jupyter Notebooks.

ADDITIONAL EXPERIENCE

Clinical Laboratory Internships

Tehran, 2022–2023

Rotations in haematology, microbiology, clinical chemistry, and biochemistry across hospital and diagnostic laboratory settings.

LANGUAGES

English: Professional proficiency

Persian (Farsi): Native

Italian: Basic communication